Supplementary Information for "Leakage-Free Evaluation of Layer Choice in Protein Language Models for Mutation Effect Prediction"

Supplementary Tables S1–S11 are numbered in the order in which they are first cited in the main text. Throughout, the primary analysis uses the full data for the wild-type arms and, under the position-based splits (modulo, contiguous, position-grouped), only single-substitution variants for the mutant arm, whose rows of a multi-site variant would share one embedding and one label across folds (main text, Section 2.2).

Assay provenance and per-dataset summary

All experiments use the 63-assay substitution snapshot distributed as LMDB files with the SaProt repository. Of these 63 assay identifiers, 46 are unchanged in ProteinGym v1.3, 13 correspond to renamed or reprocessed v1.3 assays, and 4 are no longer part of v1.3. The renamed assays and their v1.3 identifiers are listed below. Table S1 gives, for every dataset, its v1.3 status, whether the 10,000-variant cap applied, the number of analyzed rows (5,000-row ceiling), the number of analyzed rows that come from multi-site variants (whether or not other rows of the same variant survived the row subsample), the number of test rows under the modulo split whose variant also has a row in the training folds, and the mutant-arm probing result on single-substitution variants under the modulo split. In total, 11 datasets were variant-capped, 26 reach the row ceiling, and multi-site variants occur in 10 datasets, contributing 49,131 of the 213,393 analyzed rows (23.0%).

Snapshot identifier	ProteinGym v1.3 identifier
A4_HUMAN_Seuma_2021	A4_HUMAN_Seuma_2022
CP2C9_HUMAN_Amorosi_abundance_2021	CP2C9_HUMAN_Amorosi_2021_abundance
CP2C9_HUMAN_Amorosi_activity_2021	CP2C9_HUMAN_Amorosi_2021_activity
DYR_ECOLI_Thompson_plusLon_2019	DYR_ECOLI_Thompson_2019
GCN4_YEAST_Staller_induction_2018	GCN4_YEAST_Staller_2018
MTH3_HAEAE_Rockah-Shmuel_2015	MTH3_HAEAE_RockahShmuel_2015
P53_HUMAN_Giacomelli_NULL_Etoposide_2018	P53_HUMAN_Giacomelli_2018_Null_Etoposide
P53_HUMAN_Giacomelli_NULL_Nutlin_2018	P53_HUMAN_Giacomelli_2018_Null_Nutlin
P53_HUMAN_Giacomelli_WT_Nutlin_2018	P53_HUMAN_Giacomelli_2018_WT_Nutlin
SRC_HUMAN_Ahler_CD_2019	SRC_HUMAN_Ahler_2019
TPOR_HUMAN_Bridgford_S505N_2020	TPOR_HUMAN_Bridgford_2020
VKOR1_HUMAN_Chiasson_abundance_2020	VKOR1_HUMAN_Chiasson_2020_abundance
VKOR1_HUMAN_Chiasson_activity_2020	VKOR1_HUMAN_Chiasson_2020_activity

Table S1: Per-dataset provenance, caps, multi-site rows, and mutant-arm probing on single-substitution variants (modulo split, combined features): best layer, best-layer ρ, and final-layer (L32) ρ. *Shared* counts test rows under the modulo split whose variant also has a row in the training folds; *Single rows* is the number of single-substitution rows analyzed.

Dataset	v1.3	Var. cap	Rows	Multi-site rows	Shared (modulo)	Single rows	ESM-2 best L	ESM-2 best ρ	ESM-2 L32 ρ	SaProt (3Di) best L	SaProt (3Di) best ρ	SaProt (3Di) L32 ρ
A0A1I9GEU1_NEIME_Kennouche_2019	same	no	922	0	0	922	0	0.087	-0.077	3	0.004	-0.006
A0A2Z5U3Z0_9INFA_Doud_2016	same	yes	5000	0	0	5000	0	0.442	0.415	23	0.457	0.361
A4GRB6_PSEAI_Chen_2020	same	no	5000	0	0	5000	32	0.548	0.548	31	0.751	0.677
A4_HUMAN_Seuma_2021	renamed	yes	5000	4917	1004	468	26	0.565	0.475	0	0.508	0.465
AACC1_PSEAI_Dandage_2018	same	no	1801	0	0	1801	29	0.470	0.426	32	0.549	0.549
ADRB2_HUMAN_Jones_2020	same	no	5000	0	0	5000	31	0.507	0.506	31	0.527	0.526
AMIE_PSEAE_Wrenbeck_2017	same	no	5000	0	0	5000	31	0.574	0.519	31	0.640	0.631
B3VI55_LIPST_Klesmith_2015	removed	no	5000	0	0	5000	31	0.562	0.477	32	0.592	0.592
BLAT_ECOLX_Deng_2012	same	no	4996	0	0	4996	29	0.417	0.338	29	0.551	0.544
BLAT_ECOLX_Firnberg_2014	same	no	4783	0	0	4783	31	0.656	0.600	30	0.799	0.768
BLAT_ECOLX_Jacquier_2013	same	no	989	0	0	989	31	0.581	0.559	29	0.739	0.729
BLAT_ECOLX_Stiffler_2015	same	no	4996	0	0	4996	31	0.668	0.630	31	0.783	0.720
CALM1_HUMAN_Weile_2017	same	no	1813	0	0	1813	30	0.154	0.132	6	0.141	0.126
CCDB_ECOLI_Adkar_2012	same	no	1176	0	0	1176	30	0.613	0.580	30	0.694	0.639
CCDB_ECOLI_Tripathi_2016	same	no	1663	0	0	1663	30	0.459	0.443	30	0.546	0.442
CP2C9_HUMAN_Amorosi_abundance_2021	renamed	no	5000	0	0	5000	32	0.555	0.555	32	0.671	0.671
CP2C9_HUMAN_Amorosi_activity_2021	renamed	no	5000	0	0	5000	31	0.598	0.583	29	0.631	0.601
DLG4_HUMAN_Faure_2021	same	no	5000	4489	1454	1280	30	0.501	0.400	32	0.618	0.618
DLG4_RAT_McLaughlin_2012	same	no	1576	0	0	1576	29	0.471	0.404	14	0.530	0.503
DYR_ECOLI_Thompson_plusLon_2019	renamed	no	2363	0	0	2363	30	0.371	0.342	25	0.455	0.407
ESTA_BACSU_Nutschel_2020	same	no	2172	0	0	2172	17	0.349	0.320	32	0.478	0.478
F7YBW8_MESOW_Aakre_2015	same	no	5000	4995	1499	37	14	-0.464	-0.692	16	-0.482	-0.629
GAL4_YEAST_Kitzman_2015	same	no	1195	0	0	1195	30	0.471	0.451	32	0.448	0.448
GCN4_YEAST_Staller_induction_2018	renamed	no	5000	4997	4312	33	25	0.555	0.417	4	0.399	0.308
GFP_AEQVI_Sarkisyan_2016	same	yes	5000	4970	1571	1084	18	0.337	0.229	29	0.518	0.442
GRB2_HUMAN_Faure_2021	same	yes	5000	4958	1020	1034	31	0.579	0.495	32	0.648	0.648
HIS7_YEAST_Pokusaeva_2019	same	yes	5000	5000	1600	168	0	0.120	0.040	0	0.120	0.071
HSP82_YEAST_Flynn_2019	same	yes	5000	0	0	5000	29	0.453	0.449	23	0.476	0.439
HSP82_YEAST_Mishra_2016	same	no	4323	0	0	4323	27	0.507	0.465	31	0.501	0.486
IF1_ECOLI_Kelsic_2016	same	no	1367	0	0	1367	31	0.334	0.316	17	0.485	0.245
KKA2_KLEPN_Melnikov_2014	same	no	4960	0	0	4960	32	0.640	0.640	23	0.654	0.651
MK01_HUMAN_Brenan_2016	same	no	5000	0	0	5000	0	0.517	0.360	10	0.353	0.342
MSH2_HUMAN_Jia_2020	same	yes	5000	0	0	5000	31	0.521	0.519	22	0.468	0.446
MTH3_HAEAE_Rockah-Shmuel_2015	renamed	no	1777	0	0	1777	31	0.627	0.618	31	0.667	0.665
NUD15_HUMAN_Suiter_2020	same	no	2844	0	0	2844	27	0.448	0.386	31	0.674	0.651
P53_HUMAN_Giacomelli_NULL_Etoposide_2018	renamed	no	5000	0	0	5000	0	0.597	0.471	32	0.611	0.611

Dataset	v1.3	Var. cap	Rows	Multi-site rows	Shared (modulo)	Single rows	ESM-2 best L	ESM-2 best ρ	ESM-2 L32 ρ	SaProt (3Di) best L	SaProt (3Di) best ρ	SaProt (3Di) L32 ρ
P53_HUMAN_Giacomelli_NULL_Nutlin_2018	renamed	no	5000	0	0	5000	31	0.637	0.634	14	0.618	0.613
P53_HUMAN_Giacomelli_WT_Nutlin_2018	renamed	no	5000	0	0	5000	31	0.812	0.810	28	0.830	0.826
P53_HUMAN_Kotler_2018	same	no	1048	0	0	1048	31	0.684	0.594	30	0.693	0.675
P84126_THETH_Chan_2017	same	no	1519	0	0	1519	32	0.497	0.497	31	0.457	0.448
PABP_YEAST_Melamed_2013	same	yes	5000	4906	1022	1187	30	0.525	0.493	20	0.630	0.553
PTEN_HUMAN_Matreyek_2021	same	no	5000	0	0	5000	30	0.559	0.553	27	0.597	0.575
PTEN_HUMAN_Mighell_2018	same	no	5000	0	0	5000	30	0.536	0.510	31	0.539	0.529
Q59976_STRSQ_Romero_2015	same	no	2999	0	0	2999	30	0.622	0.597	32	0.665	0.665
RASH_HUMAN_Bandaru_2017	same	no	3134	0	0	3134	19	0.494	0.448	26	0.583	0.524
RL401_YEAST_Mavor_2016	removed	no	1253	0	0	1253	13	0.520	0.448	22	0.601	0.513
RL401_YEAST_Roscoe_2013	removed	no	1195	0	0	1195	30	0.603	0.532	17	0.629	0.484
RL401_YEAST_Roscoe_2014	removed	no	1380	0	0	1380	0	0.382	0.317	24	0.532	0.448
SC6A4_HUMAN_Young_2021	same	yes	5000	0	0	5000	30	0.569	0.552	32	0.569	0.569
SPG1_STRSG_Olson_2014	same	yes	5000	4993	1032	1045	12	0.508	0.470	11	0.614	0.497
SRC_HUMAN_Ahler_CD_2019	renamed	no	3372	0	0	3372	30	0.504	0.417	32	0.573	0.573
SUMO1_HUMAN_Weile_2017	same	no	1700	0	0	1700	30	0.414	0.414	32	0.447	0.447
SYUA_HUMAN_Newberry_2020	same	no	2497	0	0	2497	13	0.537	0.503	10	0.546	0.538
TADBP_HUMAN_Bolognesi_2019	same	no	1196	0	0	1196	0	0.712	0.582	0	0.689	0.679
TPK1_HUMAN_Weile_2017	same	no	3181	0	0	3181	32	0.251	0.251	19	0.220	0.199
TPMT_HUMAN_Matreyek_2018	same	no	3648	0	0	3648	30	0.552	0.502	23	0.541	0.518
TPOR_HUMAN_Bridgford_S505N_2020	renamed	no	562	0	0	562	7	0.400	0.236	12	0.481	0.387
TRPC_SACS2_Chan_2017	same	no	1519	0	0	1519	32	0.542	0.542	31	0.560	0.530
TRPC_THEMA_Chan_2017	same	no	1519	0	0	1519	32	0.358	0.358	7	0.372	0.279
UBC9_HUMAN_Weile_2017	same	no	2563	0	0	2563	30	0.392	0.360	31	0.451	0.448
VKOR1_HUMAN_Chiasson_abundance_2020	renamed	no	2695	0	0	2695	30	0.498	0.463	30	0.511	0.462
VKOR1_HUMAN_Chiasson_activity_2020	renamed	no	697	0	0	697	32	0.418	0.418	9	0.441	0.392
YAP1_HUMAN_Araya_2012	same	yes	5000	4906	1110	362	30	0.360	0.087	32	0.366	0.366

Foldseek 3Di token preparation

For each dataset, the wild-type sequence was aligned to its AlphaFold Protein Structure Database entry and each residue was assigned the Foldseek 3Di state of its aligned structural residue. Structures were obtained for 62 of 63 datasets. Two alignments were truncated at the C terminus and missing residues were padded with the Foldseek gap state (UBC9: 1 residue; Q59976: 22 residues). The influenza nucleoprotein dataset has no AFDB entry and was evaluated with structure-masked tokens in all SaProt analyses.

Table S2: 3Di files (62 datasets): token length and gap-padded residue count.

Dataset	Length	Gaps	Dataset	Length	Gaps
A0A1I9GEU1_NEIME_Kennouche_2019	161	0	MSH2_HUMAN_Jia_2020	934	0
A4GRB6_PSEAI_Chen_2020	266	0	MTH3_HAEAE_Rockah-Shmuel_2015	330	0
A4_HUMAN_Seuma_2021	770	0	NUD15_HUMAN_Suiter_2020	164	0
AACC1_PSEAI_Dandage_2018	177	0	P53_HUMAN_Giacomelli_NULL_Etoposide_2018	393	0
ADRB2_HUMAN_Jones_2020	413	0	P53_HUMAN_Giacomelli_NULL_Nutlin_2018	393	0
AMIE_PSEAE_Wrenbeck_2017	346	0	P53_HUMAN_Giacomelli_WT_Nutlin_2018	393	0
B3VI55_LIPST_Klesmith_2015	439	0	P53_HUMAN_Kotler_2018	393	0
BLAT_ECOLX_Deng_2012	286	0	P84126_THETH_Chan_2017	254	0
BLAT_ECOLX_Firnberg_2014	286	0	PABP_YEAST_Melamed_2013	577	0
BLAT_ECOLX_Jacquier_2013	286	0	PTEN_HUMAN_Matreyek_2021	403	0
BLAT_ECOLX_Stiffler_2015	286	0	PTEN_HUMAN_Mighell_2018	403	0
CALM1_HUMAN_Weile_2017	149	0	Q59976_STRSQ_Romero_2015	501	22
CCDB_ECOLI_Adkar_2012	101	0	RASH_HUMAN_Bandaru_2017	189	0
CCDB_ECOLI_Tripathi_2016	101	0	RL401_YEAST_Mavor_2016	128	0
CP2C9_HUMAN_Amorosi_abundance_2021	490	0	RL401_YEAST_Roscoe_2013	128	0
CP2C9_HUMAN_Amorosi_activity_2021	490	0	RL401_YEAST_Roscoe_2014	128	0
DLG4_HUMAN_Faure_2021	724	0	SC6A4_HUMAN_Young_2021	630	0
DLG4_RAT_McLaughlin_2012	724	0	SPG1_STRSG_Olson_2014	448	0
DYR_ECOLI_Thompson_plusLon_2019	159	0	SRC_HUMAN_Ahler_CD_2019	536	0
ESTA_BACSU_Nutschel_2020	212	0	SUMO1_HUMAN_Weile_2017	101	0
F7YBW8_MESOW_Aakre_2015	93	0	SYUA_HUMAN_Newberry_2020	140	0
GAL4_YEAST_Kitzman_2015	881	0	TADBP_HUMAN_Bolognesi_2019	414	0
GCN4_YEAST_Staller_induction_2018	281	0	TPK1_HUMAN_Weile_2017	243	0
GFP_AEQVI_Sarkisyan_2016	238	0	TPMT_HUMAN_Matreyek_2018	245	0
GRB2_HUMAN_Faure_2021	217	0	TPOR_HUMAN_Bridgford_S505N_2020	635	0
HIS7_YEAST_Pokusaeva_2019	220	0	TRPC_SACS2_Chan_2017	248	0
HSP82_YEAST_Flynn_2019	709	0	TRPC_THEMA_Chan_2017	252	0
HSP82_YEAST_Mishra_2016	709	0	UBC9_HUMAN_Weile_2017	159	1
IF1_ECOLI_Kelsic_2016	72	0	VKOR1_HUMAN_Chiasson_abundance_2020	163	0
KKA2_KLEPN_Melnikov_2014	264	0	VKOR1_HUMAN_Chiasson_activity_2020	163	0
MK01_HUMAN_Brenan_2016	360	0	YAP1_HUMAN_Araya_2012	504	0

Layer similarity: sample size and CKA estimators

Cosine similarity and linear CKA of per-layer hidden states, averaged over 10 datasets, with 200 (primary; main-text Figures 2 and 3c) or 50 randomly sampled residue positions per dataset. The standard linear CKA estimator is biased upward when the number of positions is small relative to the hidden dimension (1280); the 200-position values reduce this bias. The debiased (unbiased-HSIC) estimator is not estimable at either sample size: with fewer positions than hidden dimensions its values exceed 1, and it is listed for completeness only.

Table S3: Layer-similarity summaries.

	Adjacent layers		Layer 0 vs 32		Pearson r vs distance	
Model (positions)	cos	CKA	cos	CKA	cos	CKA
ESM-2 (200)	0.962	0.953	+0.542	0.409	-0.93	-0.66
ESM-2 (50)	0.962	0.967	+0.558	0.554	-0.93	-0.60
debiased CKA (200; not estimable)	–	1.002	–	0.963	–	–
SaProt (200)	0.979	0.969	-0.008	0.779	-0.50	-0.70
SaProt (50)	0.979	0.979	-0.009	0.808	-0.50	-0.68
debiased CKA (200; not estimable)	–	1.001	–	1.112	–	–

Best-layer versus final-layer equivalence

For each model, arm, and leakage-free split, the per-dataset paired difference between the best layer and layer 32 was computed (mutant arm on single-substitution variants; column "Data"). The best layer is the argmax of the layer-mean curve on the same per-dataset results, so selection favours finding a difference and equivalence verdicts are conservative. Verdict rule: *Equivalent* when the bootstrap 90% confidence interval of the mean paired difference (10,000 resamples, percentile method, 5th and 95th percentiles, seed 42) lies within ± 0.01; *Difference* when it lies entirely outside; *Inconclusive* otherwise. The t-based 90% interval and its verdict, the TOST p-value (margin 0.01), and Cohen's d_z are given for information: the t-based rule classifies 6 comparisons as equivalent and TOST at $\alpha = 0.05$ classifies 6, against 7 for the bootstrap rule, because several upper bounds lie close to the margin. In 6 further model–arm–split combinations the best layer is layer 32 itself (no test defined): ESM-2 masked_wt (Modulo); ESM-2 masked_wt (Contiguous); ESM-2 masked_wt (Pos.-grouped); ESM-2 unmasked_wt (Modulo); ESM-2 unmasked_wt (Contiguous); ESM-2 unmasked_wt (Pos.-grouped).

Table S4: Best-versus-final-layer comparisons, primary analysis (21 tests): 7 equivalent, 14 inconclusive, 0 difference.

Model	Arm	Split	Data	Best	n	Mean Δ	Bootstrap 90% CI	t-based 90% CI	TOST p	d_z	Verdict	t verdict
ESM-2	mutant	Modulo	single	L30	63	+0.0159	[+0.0057, +0.0269]	[+0.0050, +0.0267]	0.815	0.31	Inconclusive	Inconclusive
ESM-2	mutant	Contiguous	single	L31	57	+0.0071	[-0.0035, +0.0180]	[-0.0039, +0.0181]	0.331	0.14	Inconclusive	Inconclusive
ESM-2	mutant	Pos.-grouped	single	L30	63	+0.0187	[+0.0036, +0.0357]	[+0.0024, +0.0350]	0.812	0.24	Inconclusive	Inconclusive
SaProt (mask)	masked_wt	Modulo	full	L30	63	+0.0054	[+0.0011, +0.0098]	[+0.0009, +0.0099]	0.045	0.25	Equivalent	Equivalent
SaProt (mask)	masked_wt	Contiguous	full	L31	57	+0.0077	[+0.0034, +0.0119]	[+0.0034, +0.0120]	0.188	0.39	Inconclusive	Inconclusive
SaProt (mask)	masked_wt	Pos.-grouped	full	L31	63	+0.0035	[+0.0003, +0.0066]	[+0.0002, +0.0067]	0.001	0.22	Equivalent	Equivalent
SaProt (mask)	unmasked_wt	Modulo	full	L30	63	+0.0063	[+0.0010, +0.0118]	[+0.0007, +0.0119]	0.136	0.24	Inconclusive	Inconclusive
SaProt (mask)	unmasked_wt	Contiguous	full	L31	57	+0.0041	[-0.0013, +0.0096]	[-0.0015, +0.0097]	0.041	0.16	Equivalent	Equivalent
SaProt (mask)	unmasked_wt	Pos.-grouped	full	L27	63	+0.0031	[-0.0038, +0.0099]	[-0.0038, +0.0101]	0.052	0.09	Equivalent	Inconclusive
SaProt (mask)	mutant	Modulo	single	L30	63	+0.0050	[-0.0027, +0.0126]	[-0.0029, +0.0130]	0.150	0.13	Inconclusive	Inconclusive
SaProt (mask)	mutant	Contiguous	single	L30	57	+0.0031	[-0.0069, +0.0135]	[-0.0073, +0.0134]	0.134	0.07	Inconclusive	Inconclusive
SaProt (mask)	mutant	Pos.-grouped	single	L30	63	+0.0057	[-0.0044, +0.0161]	[-0.0047, +0.0161]	0.246	0.11	Inconclusive	Inconclusive
SaProt (3Di)	masked_wt	Modulo	full	L31	63	+0.0017	[-0.0010, +0.0046]	[-0.0012, +0.0047]	0.000	0.13	Equivalent	Equivalent
SaProt (3Di)	masked_wt	Contiguous	full	L31	57	+0.0009	[-0.0039, +0.0057]	[-0.0040, +0.0057]	0.001	0.04	Equivalent	Equivalent
SaProt (3Di)	masked_wt	Pos.-grouped	full	L31	63	+0.0016	[-0.0011, +0.0043]	[-0.0012, +0.0043]	0.000	0.12	Equivalent	Equivalent
SaProt (3Di)	unmasked_wt	Modulo	full	L31	63	+0.0071	[+0.0039, +0.0106]	[+0.0037, +0.0105]	0.081	0.44	Inconclusive	Inconclusive
SaProt (3Di)	unmasked_wt	Contiguous	full	L31	57	+0.0086	[+0.0036, +0.0138]	[+0.0033, +0.0139]	0.328	0.36	Inconclusive	Inconclusive
SaProt (3Di)	unmasked_wt	Pos.-grouped	full	L31	63	+0.0083	[+0.0045, +0.0126]	[+0.0042, +0.0125]	0.254	0.42	Inconclusive	Inconclusive
SaProt (3Di)	mutant	Modulo	single	L30	63	+0.0060	[-0.0013, +0.0133]	[-0.0016, +0.0136]	0.190	0.17	Inconclusive	Inconclusive
SaProt (3Di)	mutant	Contiguous	single	L31	57	+0.0100	[+0.0019, +0.0187]	[+0.0013, +0.0187]	0.501	0.26	Inconclusive	Inconclusive
SaProt (3Di)	mutant	Pos.-grouped	single	L31	63	+0.0088	[+0.0031, +0.0146]	[+0.0029, +0.0147]	0.368	0.32	Inconclusive	Inconclusive

Full 33-layer probing curves

Mean Spearman ρ per layer (combined features) for every arm and split; best layer per column in bold. $n = 63$ datasets ($n = 57$ for the contiguous split). Columns marked † (mutant arm, leakage-free splits) are computed on single-substitution variants.

Table S5: ESM-2 650M: per-layer mean ρ by arm and split.

	masked_wt				unmasked_wt				mutant			
Layer	Random	Modulo	Contig.	Pos.-gr.	Random	Modulo	Contig.	Pos.-gr.	Random	Modulo†	Contig.†	Pos.-gr.†
0	0.601	0.232	0.067	0.224	0.601	0.191	0.046	0.188	0.510	0.356	0.189	0.349
1	0.601	0.235	0.129	0.219	0.602	0.257	0.167	0.244	0.615	0.245	0.150	0.224
2	0.602	0.262	0.148	0.252	0.602	0.289	0.197	0.281	0.630	0.253	0.146	0.243
3	0.602	0.275	0.155	0.267	0.603	0.306	0.197	0.299	0.639	0.256	0.154	0.251
4	0.603	0.285	0.159	0.276	0.603	0.316	0.199	0.305	0.645	0.258	0.158	0.248
5	0.602	0.301	0.165	0.290	0.603	0.331	0.208	0.319	0.647	0.272	0.160	0.265
6	0.603	0.307	0.167	0.298	0.603	0.334	0.205	0.323	0.650	0.288	0.154	0.274
7	0.603	0.312	0.169	0.303	0.603	0.336	0.208	0.325	0.649	0.304	0.159	0.291
8	0.603	0.316	0.174	0.306	0.603	0.342	0.218	0.329	0.650	0.308	0.169	0.295
9	0.603	0.322	0.180	0.308	0.603	0.343	0.211	0.328	0.649	0.312	0.162	0.294
10	0.603	0.324	0.195	0.315	0.604	0.346	0.225	0.335	0.652	0.325	0.169	0.308
11	0.603	0.334	0.198	0.323	0.604	0.349	0.221	0.337	0.652	0.339	0.186	0.324
12	0.603	0.332	0.201	0.326	0.604	0.349	0.217	0.338	0.656	0.330	0.188	0.320
13	0.603	0.339	0.204	0.327	0.603	0.350	0.218	0.337	0.659	0.313	0.163	0.312
14	0.603	0.336	0.205	0.328	0.603	0.345	0.224	0.338	0.658	0.296	0.149	0.296
15	0.603	0.340	0.203	0.331	0.603	0.351	0.220	0.342	0.658	0.292	0.129	0.295
16	0.603	0.344	0.199	0.333	0.604	0.356	0.218	0.347	0.660	0.299	0.131	0.292
17	0.604	0.348	0.196	0.341	0.604	0.361	0.211	0.354	0.660	0.307	0.132	0.296
18	0.604	0.347	0.190	0.339	0.604	0.359	0.205	0.351	0.660	0.307	0.132	0.295
19	0.604	0.350	0.206	0.341	0.604	0.361	0.218	0.354	0.660	0.308	0.144	0.295
20	0.604	0.359	0.210	0.352	0.604	0.368	0.228	0.363	0.660	0.306	0.133	0.300
21	0.604	0.364	0.213	0.354	0.604	0.374	0.230	0.365	0.660	0.300	0.132	0.290
22	0.604	0.364	0.215	0.355	0.604	0.374	0.235	0.367	0.661	0.294	0.147	0.283
23	0.604	0.368	0.228	0.361	0.604	0.381	0.247	0.372	0.662	0.302	0.162	0.291
24	0.604	0.377	0.236	0.369	0.604	0.390	0.256	0.380	0.662	0.310	0.152	0.294
25	0.604	0.387	0.248	0.378	0.604	0.398	0.264	0.389	0.664	0.314	0.150	0.302
26	0.604	0.398	0.268	0.389	0.605	0.412	0.285	0.403	0.666	0.327	0.170	0.323
27	0.605	0.406	0.281	0.399	0.605	0.422	0.300	0.414	0.670	0.350	0.200	0.349
28	0.605	0.424	0.313	0.414	0.605	0.437	0.329	0.428	0.673	0.378	0.247	0.376
29	0.605	0.435	0.327	0.426	0.606	0.448	0.345	0.440	0.676	0.408	0.282	0.409
30	**0.606**	0.449	0.350	0.440	**0.606**	0.463	0.365	0.453	0.679	**0.443**	0.336	**0.444**
31	**0.606**	0.453	0.365	0.446	**0.606**	0.465	0.376	0.457	0.679	0.439	**0.353**	0.435
32	**0.606**	**0.459**	**0.370**	**0.450**	**0.606**	**0.470**	**0.388**	**0.461**	**0.681**	0.427	0.346	0.425

Table S6: SaProt (structure-masked): per-layer mean ρ by arm and split.

	masked_wt				unmasked_wt				mutant			
Layer	Random	Modulo	Contig.	Pos.-gr.	Random	Modulo	Contig.	Pos.-gr.	Random	Modulo†	Contig.†	Pos.-gr.†
0	0.600	0.241	0.048	0.229	0.601	0.234	0.119	0.227	0.537	0.339	0.175	0.322
1	0.602	0.230	0.137	0.221	0.602	0.262	0.191	0.250	0.611	0.254	0.172	0.245
2	0.602	0.250	0.151	0.246	0.602	0.292	0.205	0.286	0.617	0.274	0.171	0.263
3	0.602	0.262	0.166	0.255	0.602	0.307	0.218	0.303	0.620	0.276	0.175	0.263
4	0.602	0.285	0.176	0.275	0.602	0.328	0.219	0.319	0.628	0.289	0.166	0.276
5	0.602	0.292	0.172	0.285	0.603	0.329	0.222	0.322	0.630	0.294	0.175	0.284
6	0.602	0.307	0.174	0.300	0.603	0.339	0.224	0.332	0.636	0.309	0.184	0.299
7	**0.603**	0.317	0.178	0.311	0.603	0.349	0.231	0.343	0.637	0.318	0.184	0.304
8	**0.603**	0.335	0.197	0.327	0.603	0.362	0.233	0.356	0.642	0.342	0.195	0.330
9	**0.603**	0.339	0.200	0.332	0.603	0.368	0.239	0.362	0.646	0.349	0.197	0.338
10	0.603	0.342	0.193	0.335	0.603	0.369	0.234	0.363	0.647	0.350	0.198	0.338
11	0.603	0.352	0.205	0.344	0.603	0.374	0.243	0.368	0.648	0.360	0.211	0.349
12	0.602	0.358	0.212	0.350	0.603	0.383	0.253	0.375	0.647	0.368	0.213	0.356
13	0.602	0.359	0.219	0.353	0.603	0.386	0.260	0.379	0.648	0.372	0.232	0.359
14	0.602	0.364	0.229	0.360	0.603	0.391	0.266	0.383	0.649	0.378	0.247	0.368
15	0.602	0.366	0.233	0.362	0.603	0.396	0.273	0.387	0.649	0.385	0.244	0.373
16	0.602	0.370	0.234	0.364	0.603	0.397	0.270	0.386	0.650	0.389	0.253	0.375
17	0.602	0.371	0.240	0.367	0.603	0.400	0.274	0.389	0.650	0.393	0.262	0.379
18	0.603	0.374	0.247	0.368	0.603	0.401	0.280	0.389	0.649	0.401	0.264	0.384
19	0.602	0.374	0.250	0.368	0.603	0.401	0.281	0.389	0.649	0.400	0.267	0.389
20	0.602	0.374	0.249	0.370	0.603	0.403	0.282	0.392	0.649	0.398	0.272	0.390
21	0.602	0.374	0.244	0.370	0.603	0.405	0.288	0.395	0.650	0.404	0.273	0.394
22	0.602	0.377	0.249	0.373	0.603	0.408	0.291	0.398	0.651	0.411	0.278	0.403
23	0.602	0.381	0.253	0.377	**0.603**	0.409	0.290	0.400	**0.652**	0.418	0.283	0.409
24	0.602	0.379	0.254	0.376	**0.603**	0.406	0.292	0.399	0.650	0.417	0.283	0.413
25	0.602	0.378	0.252	0.374	**0.603**	0.408	0.298	0.400	0.650	0.412	0.284	0.405
26	0.602	0.383	0.256	0.379	**0.603**	0.410	0.302	0.402	0.651	0.419	0.288	0.411
27	0.603	0.387	0.261	0.384	**0.604**	0.411	0.305	**0.403**	0.650	0.423	0.296	0.418
28	0.603	0.389	0.266	0.383	**0.604**	0.411	0.309	0.401	0.650	0.430	0.300	0.423
29	**0.603**	0.388	0.271	0.383	**0.604**	**0.412**	0.311	0.401	0.650	0.428	0.304	0.424
30	**0.603**	**0.391**	0.277	0.383	**0.604**	**0.412**	0.317	0.401	0.650	**0.434**	**0.310**	**0.429**
31	**0.603**	0.391	**0.282**	**0.385**	**0.604**	0.411	**0.324**	**0.403**	0.650	0.429	0.309	0.425
32	**0.603**	0.386	0.274	0.382	**0.604**	0.406	0.320	0.400	0.651	0.428	0.307	0.423

Table S7: SaProt (real 3Di): per-layer mean ρ by arm and split.

	masked_wt				unmasked_wt				mutant			
Layer	Random	Modulo	Contig.	Pos.-gr.	Random	Modulo	Contig.	Pos.-gr.	Random	Modulo†	Contig.†	Pos.-gr.†
0	0.601	0.274	0.112	0.251	0.602	0.276	0.191	0.263	0.578	0.317	0.206	0.314
1	0.603	0.299	0.188	0.289	0.603	0.333	0.238	0.314	0.629	0.329	0.226	0.319
2	0.603	0.337	0.221	0.326	0.604	0.362	0.255	0.346	0.640	0.352	0.251	0.347
3	0.603	0.346	0.233	0.336	0.604	0.376	0.268	0.361	0.649	0.365	0.253	0.357
4	0.603	0.364	0.250	0.357	0.604	0.384	0.276	0.373	0.653	0.370	0.254	0.364
5	0.604	0.372	0.264	0.365	0.604	0.393	0.291	0.383	0.656	0.374	0.273	0.368
6	0.604	0.379	0.271	0.372	0.604	0.396	0.294	0.385	0.660	0.380	0.277	0.375
7	0.604	0.387	0.272	0.378	0.604	0.402	0.295	0.393	0.662	0.380	0.271	0.372
8	0.604	0.398	0.284	0.390	0.604	0.410	0.295	0.401	0.665	0.395	0.264	0.382
9	0.604	0.398	0.290	0.392	0.604	0.413	0.301	0.404	0.667	0.398	0.268	0.388
10	0.604	0.399	0.288	0.393	0.605	0.415	0.303	0.406	0.669	0.391	0.269	0.380
11	0.604	0.407	0.298	0.399	0.605	0.422	0.312	0.412	0.671	0.403	0.273	0.392
12	0.605	0.411	0.306	0.404	0.605	0.427	0.321	0.417	0.671	0.408	0.290	0.396
13	0.604	0.416	0.311	0.408	0.605	0.431	0.325	0.422	0.671	0.410	0.297	0.394
14	0.605	0.420	0.317	0.413	0.605	0.436	0.335	0.428	0.671	0.415	0.306	0.402
15	0.605	0.421	0.321	0.414	0.605	0.439	0.341	0.430	0.672	0.422	0.315	0.410
16	0.605	0.424	0.326	0.417	0.605	0.443	0.348	0.435	0.672	0.430	0.327	0.419
17	0.605	0.426	0.332	0.420	0.605	0.444	0.352	0.436	0.674	0.432	0.327	0.424
18	0.605	0.427	0.335	0.421	0.605	0.447	0.358	0.439	0.674	0.430	0.332	0.427
19	0.605	0.428	0.336	0.422	**0.605**	0.448	0.360	0.440	0.675	0.434	0.341	0.427
20	0.605	0.431	0.343	0.425	**0.605**	0.451	0.367	0.443	0.677	0.439	0.355	0.433
21	0.605	0.433	0.345	0.429	**0.605**	0.454	0.370	0.446	0.678	0.440	0.357	0.437
22	**0.605**	0.436	0.352	0.432	**0.606**	0.456	0.376	0.449	0.679	0.458	0.371	0.448
23	**0.605**	0.439	0.358	0.434	**0.606**	0.458	0.382	0.451	0.681	0.459	0.373	0.448
24	**0.605**	0.442	0.365	0.437	**0.606**	0.460	0.386	0.453	0.682	0.461	0.377	0.454
25	**0.605**	0.443	0.365	0.438	**0.606**	0.462	0.387	0.454	0.683	0.464	0.380	0.457
26	**0.605**	0.447	0.369	0.441	**0.606**	0.464	0.392	0.457	**0.684**	0.468	0.391	0.466
27	**0.605**	0.452	0.375	0.445	**0.606**	0.468	0.398	0.460	**0.684**	0.470	0.399	0.470
28	**0.605**	0.455	0.379	0.448	**0.606**	0.469	0.399	0.461	**0.684**	0.482	0.422	0.482
29	**0.605**	0.456	0.385	0.450	**0.605**	0.469	0.403	0.463	**0.684**	0.483	0.423	0.481
30	**0.605**	0.459	0.392	**0.454**	**0.605**	0.471	0.410	**0.467**	0.682	**0.492**	0.445	**0.490**
31	**0.606**	**0.461**	**0.394**	**0.454**	**0.605**	**0.473**	**0.415**	**0.467**	0.684	0.491	**0.449**	**0.490**
32	**0.606**	0.460	0.394	0.453	**0.605**	0.466	0.406	0.459	0.683	0.486	0.439	0.481

Single-substitution analysis of all arms

Probing repeated for all three model configurations on single-substitution variants only (one row per variant; variant cap applied after filtering) under the leakage-free splits, with the same folds and readout. For comparison, the last two columns give the best-layer ρ on the full data, in which rows of a multi-site variant can share a label (all arms) and an embedding (mutant arm) across folds. Δ is the mean per-dataset difference between the best layer and layer 32 on single substitutions, with its bootstrap 90% confidence interval and the verdict under the rule of Table S4 (no test when the best layer is layer 32). On single substitutions, 3 comparisons are equivalent, 20 inconclusive, and 0 indicate a difference beyond the margin; the largest point estimate is 0.019 and the largest 90% upper bound 0.036. Two datasets retain fewer than 50 single-substitution variants (F7YBW8_MESOW_Aakre_2015, 37; GCN4_YEAST_Staller_induction_2018, 33; Table S1); excluding them, the best layer remains in layers 30–32 in every column and the largest point estimate and 90% upper bound become 0.014 and 0.025.

Table S8: Single-substitution probing versus full data, leakage-free splits (combined features).

			Single substitutions						Full data	
Model	Arm	Split	n	Best	Best ρ	L32 ρ	Δ (90% CI)	Verdict	Best	Best ρ
ESM-2	masked_wt	Modulo	63	L31	0.476	0.473	+0.0024 [-0.010, +0.022]	Inconclusive	L32	0.459
ESM-2	masked_wt	Contiguous	57	L32	0.386	0.386	–	–	L32	0.370
ESM-2	masked_wt	Pos.-grouped	63	L32	0.466	0.466	–	–	L32	0.450
ESM-2	unmasked_wt	Modulo	63	L31	0.492	0.486	+0.0060 [-0.008, +0.029]	Inconclusive	L32	0.470
ESM-2	unmasked_wt	Contiguous	57	L32	0.404	0.404	–	–	L32	0.388
ESM-2	unmasked_wt	Pos.-grouped	63	L32	0.475	0.475	–	–	L32	0.461
ESM-2	mutant	Modulo	63	L30	0.443	0.427	+0.0159 [+0.006, +0.027]	Inconclusive	L31	0.483
ESM-2	mutant	Contiguous	57	L31	0.353	0.346	+0.0071 [-0.003, +0.018]	Inconclusive	L31	0.365
ESM-2	mutant	Pos.-grouped	63	L30	0.444	0.425	+0.0187 [+0.004, +0.036]	Inconclusive	L31	0.477
SaProt (mask)	masked_wt	Modulo	63	L31	0.417	0.412	+0.0045 [+0.000, +0.009]	Equivalent	L30	0.391
SaProt (mask)	masked_wt	Contiguous	57	L31	0.290	0.283	+0.0069 [+0.002, +0.011]	Inconclusive	L31	0.282
SaProt (mask)	masked_wt	Pos.-grouped	63	L31	0.401	0.398	+0.0027 [-0.001, +0.006]	Equivalent	L31	0.385
SaProt (mask)	unmasked_wt	Modulo	63	L30	0.438	0.429	+0.0083 [+0.001, +0.016]	Inconclusive	L30	0.412
SaProt (mask)	unmasked_wt	Contiguous	57	L31	0.339	0.333	+0.0054 [+0.000, +0.011]	Inconclusive	L31	0.324
SaProt (mask)	unmasked_wt	Pos.-grouped	63	L27	0.426	0.417	+0.0090 [-0.003, +0.024]	Inconclusive	L27	0.403
SaProt (mask)	mutant	Modulo	63	L30	0.434	0.428	+0.0050 [-0.003, +0.013]	Inconclusive	L30	0.465
SaProt (mask)	mutant	Contiguous	57	L30	0.310	0.307	+0.0031 [-0.007, +0.013]	Inconclusive	L32	0.323
SaProt (mask)	mutant	Pos.-grouped	63	L30	0.429	0.423	+0.0057 [-0.004, +0.016]	Inconclusive	L30	0.461
SaProt (3Di)	masked_wt	Modulo	63	L31	0.484	0.475	+0.0096 [-0.002, +0.029]	Inconclusive	L31	0.461
SaProt (3Di)	masked_wt	Contiguous	57	L30	0.406	0.404	+0.0017 [-0.006, +0.011]	Inconclusive	L31	0.394
SaProt (3Di)	masked_wt	Pos.-grouped	63	L31	0.470	0.470	+0.0001 [-0.004, +0.004]	Equivalent	L31	0.454
SaProt (3Di)	unmasked_wt	Modulo	63	L31	0.500	0.493	+0.0067 [+0.003, +0.010]	Inconclusive	L31	0.473
SaProt (3Di)	unmasked_wt	Contiguous	57	L31	0.431	0.423	+0.0075 [+0.002, +0.013]	Inconclusive	L31	0.415
SaProt (3Di)	unmasked_wt	Pos.-grouped	63	L30	0.491	0.479	+0.0123 [+0.004, +0.024]	Inconclusive	L31	0.467
SaProt (3Di)	mutant	Modulo	63	L30	0.492	0.486	+0.0060 [-0.001, +0.013]	Inconclusive	L31	0.531
SaProt (3Di)	mutant	Contiguous	57	L31	0.449	0.439	+0.0100 [+0.002, +0.019]	Inconclusive	L31	0.451
SaProt (3Di)	mutant	Pos.-grouped	63	L31	0.490	0.481	+0.0088 [+0.003, +0.015]	Inconclusive	L31	0.532

Multi-scale pooling: paired comparisons

Paired per-dataset differences (strategy minus last layer) for the pooling benchmark of main Table 3 (mutant arm, combined features, 20 datasets per model; SaProt with structure-masked inputs) under the modulo split, under the random split, and under the modulo split after excluding the two datasets that contain multi-site variants (A4_HUMAN_Seuma_2021, DLG4_HUMAN_Faure_2021; 18 datasets). Entries give the mean difference with its bootstrap 90% confidence interval; † marks intervals excluding zero. Under the modulo split no combination significantly improves on the last layer, with or without the two multi-site datasets; under the random split several fusion strategies do.

Table S9: Paired fusion-minus-last-layer differences, $\Delta\rho$ (90% CI).

		ESM-2			SaProt		
Readout	Strategy	Modulo	Random	Modulo, excl.	Modulo	Random	Modulo, excl.
Ridge	Mean L20–33	+0.010 (-0.001, +0.023)	-0.001 (-0.004, +0.002)	+0.011 (-0.001, +0.025)	+0.001 (-0.012, +0.014)	+0.003 (-0.001, +0.007)	+0.001 (-0.013, +0.015)
Ridge	Concat 5L	+0.018 (-0.008, +0.044)	+0.012 (+0.005, +0.021)†	+0.009 (-0.017, +0.035)	+0.003 (-0.016, +0.021)	+0.030 (+0.020, +0.039)†	-0.004 (-0.023, +0.016)
Ridge	Attention	-0.024 (-0.053, +0.002)	-0.002 (-0.011, +0.007)	-0.033 (-0.063, -0.006)†	-0.002 (-0.012, +0.007)	+0.010 (+0.004, +0.016)†	-0.004 (-0.014, +0.007)
RF	Mean L20–33	-0.056 (-0.070, -0.042)†	-0.002 (-0.007, +0.004)	-0.060 (-0.074, -0.046)†	+0.000 (-0.009, +0.011)	+0.038 (+0.023, +0.053)†	+0.002 (-0.009, +0.013)
RF	Concat 5L	-0.023 (-0.031, -0.016)†	+0.009 (+0.005, +0.013)†	-0.025 (-0.033, -0.017)†	-0.008 (-0.016, +0.000)	+0.064 (+0.041, +0.089)†	-0.010 (-0.018, -0.001)†
RF	Attention	-0.057 (-0.099, -0.023)†	-0.006 (-0.018, +0.005)	-0.061 (-0.106, -0.024)†	-0.009 (-0.019, +0.001)	+0.059 (+0.036, +0.084)†	-0.010 (-0.021, +0.002)
MLP	Mean L20–33	-0.044 (-0.064, -0.024)†	-0.004 (-0.010, +0.002)	-0.050 (-0.072, -0.029)†	-0.016 (-0.029, -0.003)†	+0.017 (+0.009, +0.025)†	-0.018 (-0.031, -0.005)†
MLP	Concat 5L	-0.045 (-0.069, -0.020)†	-0.019 (-0.030, -0.008)†	-0.054 (-0.078, -0.027)†	-0.039 (-0.057, -0.022)†	+0.002 (-0.013, +0.015)	-0.043 (-0.061, -0.026)†
MLP	Attention	-0.040 (-0.064, -0.015)†	+0.007 (-0.008, +0.023)	-0.055 (-0.075, -0.036)†	-0.019 (-0.029, -0.009)†	+0.026 (+0.017, +0.035)†	-0.022 (-0.032, -0.011)†
LightGBM	Mean L20–33	-0.050 (-0.065, -0.035)†	-0.002 (-0.005, +0.002)	-0.055 (-0.070, -0.038)†	-0.009 (-0.018, +0.001)	+0.030 (+0.019, +0.040)†	-0.010 (-0.021, -0.000)†
LightGBM	Concat 5L	-0.013 (-0.025, -0.002)†	+0.009 (+0.005, +0.012)†	-0.017 (-0.030, -0.006)†	-0.012 (-0.018, -0.006)†	+0.050 (+0.033, +0.067)†	-0.014 (-0.020, -0.009)†
LightGBM	Attention	-0.045 (-0.086, -0.011)†	-0.002 (-0.010, +0.005)	-0.051 (-0.094, -0.014)†	-0.014 (-0.027, -0.002)†	+0.046 (+0.029, +0.063)†	-0.018 (-0.032, -0.005)†

Last-layer readout contrast by split

Paired last-layer LightGBM minus Ridge difference over the 20 pooling datasets.

Table S10: Last-layer readout contrast, LightGBM minus Ridge.

Model	Split	Mean $\Delta\rho$	90% CI	Wilcoxon p
ESM-2	Random	-0.035	(-0.044, -0.027)	1.3e-05
ESM-2	Modulo	+0.045	(+0.021, +0.068)	0.011
ESM-2	Contiguous	+0.058	(+0.026, +0.090)	0.013
SaProt	Random	-0.065	(-0.080, -0.050)	9.5e-06
SaProt	Modulo	+0.016	(-0.016, +0.048)	0.57
SaProt	Contiguous	+0.053	(+0.006, +0.102)	0.12

Published ProteinGym anchors

Published v1.3 results recomputed on the 46 assays shared with this work: zero-shot values are the per-assay Spearman correlations of the official v1.3 zero-shot benchmark table; supervised values are computed from the out-of-fold predictions that the v1.3 release provides for published supervised models under its official fold schemes. Of the shared assays, 38 contain single substitutions only; on these the final-layer logit lens reproduces the published values, and the residual differences on the full set arise in multi-site assays, where the lens scores each substitution of a variant separately rather than summing over the variant.

Table S11: Published ProteinGym v1.3 results recomputed on the 46 assays shared with this work. (A) Zero-shot: mean per-assay Spearman ρ on all shared assays and on the 38 shared assays containing single substitutions only. (B) Supervised: official out-of-fold predictions under the v1.3 fold schemes; our rows give the final-layer mutant-arm probing (combined features) on single-substitution variants of the same assays, computed on the capped snapshot data ($n = 41$ for the contiguous split).

A. Zero-shot

Method	All shared ($n = 46$)	Single-substitution ($n = 38$)
ESM-2 650M (published)	0.472	0.476
SaProt 650M (published)	0.481	0.477
ESM-2 150M (published)	0.421	0.422
ESM-2 3B (published)	0.448	0.454
ESM-1v single (published)	0.445	0.458
MSA Transformer single (published)	0.447	0.457
GEMME (published)	0.474	0.468
EVE single (published)	0.478	0.470
This work: lens, ESM-2 final layer	0.438	0.475
This work: lens, SaProt (3Di) final layer	0.440	0.473
This work: SaProt zero-shot reproduction	0.484	0.478

B. Supervised ($n = 46$)

Method	Random	Modulo	Contiguous
Kermut	0.711	0.605	0.561
ProteinNPT	0.697	0.566	0.525
Emb.-aug. MSA Transformer	0.640	0.572	0.536
Emb.-aug. ESM-1v	0.612	0.539	0.504
OHE-aug. MSA Transformer	0.546	0.459	0.456
OHE, not augmented	0.570	-0.003	0.027
This work: ESM-2 final layer	0.676	0.407	0.338
This work: SaProt (3Di) final layer	0.680	0.469	0.431